

Stacked Hexagonal Magnetic Columns with Field-Coupled Generator–Sensor Loops: A Simulation and Micromagnetic Study of In-Memory Reservoir Computing and Associative Memory

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Abstract

We propose and evaluate in simulation a three-dimensional magnetic computing architecture built from hexagonal columns of stacked layers. Perpendicular nanomagnet generators and magnetic field sensors sit on the column walls and in a central pillar, and each generator is driven by a weighted combination of sensors that read the superposed stray field of nearby generators, so the field superposition performs the weighted summation of a recurrent network while the magnetization stores its state. Among four placements, a layout that alternates generators between the walls and the pillar from layer to layer (variant C) performs best. MuMax3 simulations of 30 nm perpendicular magnets reproduce stray-field superposition with a 3.1% median error, show that generators stay saturated, and show that the near field is about half of the point-dipole value. With the coupling calibrated to MuMax3, the best configuration reaches a 5-step Mackey–Glass prediction error of 0.091 ± 0.008 , against 0.16 for an echo state network with the same input nodes, and recalls 24 random patterns with an overlap of 0.95, although this capacity relies mainly on 64 addressed links per generator and field coupling alone does not connect cells. As an optional alternative to direct field sensors, we also examined a skyrmion film: a skyrmion follows the summed field of several generators (correlation 0.998), adds an intrinsic nonlinearity and, confined in an 80 nm disk with a 2 nm magnetic layer, reads the generator state correctly at 300 K in 18 of 18 one-nanosecond windows, whereas writing skyrmions by current or gate voltage was not practical in our tests.

Keywords: nanomagnetic computing, physical reservoir computing, in-memory computing, associative memory, stray-field coupling, three-dimensional integration, micromagnetics

1. Introduction

Moving data between memory and processing units dominates the energy cost of many machine-learning workloads, which has motivated a broad search for physical substrates that compute where the data are stored [4, 14, 16]. Magnetic systems are attractive for this purpose because a nanomagnet is non-volatile, switches in nanoseconds, and interacts with its neighbours through its stray field without wires. Field-coupled nanomagnetic logic has demonstrated Boolean gates, including gates that couple magnets in different layers of a three-dimensional stack [6, 7, 8], and arrays of dipole-coupled nanomagnets, spin-torque oscillators and artificial spin ices have been used as physical reservoirs whose state is read out by a trained linear layer [5, 9, 10, 11, 12, 13].

This work grew out of the development of a neural model for assessing companies, intended for integration into the VeritasLinks platform. The underlying idea was to give a neural network a physical body: a model whose logical structure is the same as the physical structure that runs it, so that the network can reorganise its connections and its memory at both levels, in software by reprogramming its weights and in hardware by changing the state of the elements that carry them, and in this way optimise itself. Such a network keeps and remembers its own state between observations instead of recomputing each assessment from scratch, because memory and computation are the same physical state. This paper studies a first physical substrate for that idea and does not evaluate companies. Here the reorganisation is limited to programming the weights from sensors to generators; self-reorganisation of the physical structure itself is a longer-term aim and is not demonstrated.

The architectural question studied here is the following. If generators (switchable nanomagnets) and sensors (field-sensitive readers) are placed on the walls of vertical hexagonal columns and coupled into

closed loops, so that each generator is driven by the field its sensors measure, what kind of computation does the column support, how much information can it store, and which geometric choices matter? The idea is that the superposition of stray fields inside a column performs the weighted summation of a recurrent network, the magnetization state holds the network state, and a central pillar running through each column integrates the field of the whole column.

We built a dynamical model of this architecture with a voxel heat model, evaluated it on standard reservoir-computing and associative-memory benchmarks against strong baselines, and then checked the physical assumptions of the coupling with micromagnetic simulations in MuMax3 [22]. The contributions are:

- (i) A column architecture with generator–sensor loops on hexagonal walls and a central pillar, and a comparison of four placements of generators and sensors, of which an alternating-layer placement (variant C) performs best.
- (ii) A micromagnetic validation of stray-field superposition for perpendicular nanomagnets, a sweep of 144 configurations up to 19 cells and 10 layers, and a coupling law calibrated to MuMax3 that replaces the point-dipole approximation.
- (iii) Benchmarks on held-out random seeds with 95% confidence intervals against tuned echo state networks, including results where the architecture is weaker than the baseline.
- (iv) Negative results with their mechanisms: field coupling alone does not carry signals between cells, the inter-cell logic shows no regime between silence and self-sustained activity, and associative capacity is bounded by the number of programmable parameters per generator.
- (v) An optional alternative sensor: a skyrmion film examined in MuMax3 at 0 K and at 300 K, including a confinement geometry in which it reads the generator state at room temperature, and tests of writing skyrmions by current and by gate voltage.

2. Related work

Nanomagnetic logic. Field-coupled nanomagnets implement majority gates whose output follows the summed coupling field of the inputs [6]. In perpendicular nanomagnetic logic each input is weighted by its geometry and its distance to the output, which turns a gate into a threshold element [8], and a three-dimensional majority gate has been demonstrated with one input magnet 60 nm below the others [7]. The present architecture uses the same physical summation but organises it into closed sensor–generator loops along vertical columns rather than into feed-forward gates.

Nanomagnetic reservoirs. Nomura et al. proposed reservoir computing with arrays of dipole-coupled nanomagnets compatible with magnetoresistive memory integration [9]. Edwards et al. used frustrated, passively coupled nanomagnets read by magnetic tunnel junctions with a trained memristor output layer [10], and Gartside et al. trained reservoirs on an artificial spin-vortex ice through its spin-wave spectrum [11]. Spin-torque nano-oscillators have been used both as single-device reservoirs and as coupled networks [12, 13]. Magnetic skyrmions have been proposed as reservoir elements, both as a single non-linear resistive element [28] and as random skyrmion textures [27], and a perspective on unconventional computing with skyrmions also surveys a range of physical reservoir approaches [26]. A review of nanomagnetic reservoirs is given in [5], and open tools such as PRCpy [29] make the processing of physical reservoir data easier to compare across substrates. In these systems the coupling between elements is passive and fixed by the geometry; in the architecture studied here the generators are actively driven by sensor readings, and the weights from sensors to generators can be fixed or programmed.

Magnetic neurons and memory. Domain-wall devices have been used as neurons with lateral inhibition through magnetostatic interaction [15], magnetoresistive crossbars perform in-memory multiply–accumulate operations [16], and associative memories follow the Hopfield model [17] whose capacity is about 0.138 N patterns for N neurons [18]. A single perceptron with N inputs can store up to $2N$ random associations [19], which sets the scale for the per-generator capacity limits discussed in Section 7.

Thermal management. Embedded microfluidic cooling has been co-designed with electronics at the micrometre scale [25]. At the 75 nm gaps considered here fluid channels are not realistic, and the corner channels of our design are modelled as solid heat sinks rather than as fluidic paths.

3. Architecture

The network is a hexagonal lattice of cells (Fig. 1a): one cell, a ring of 7, or two rings totalling 19. Each cell is a vertical column of L layers. On every layer each of the six walls carries two element positions at one third and two thirds of its length, and a central pillar of radius r_c carries one position per face, giving 12 wall positions and 6 pillar positions per layer. An element position holds either a generator, a nanomagnet whose magnetization $s \in [-1, 1]$ is set by a local drive, or a sensor that measures the component of the magnetic field along the generator axis. Generators are perpendicular (axial) magnets unless stated otherwise. Neighbouring cells are separated by a gap, and at every point where three cells meet a corner channel connects the nearest generators of the adjacent cells.

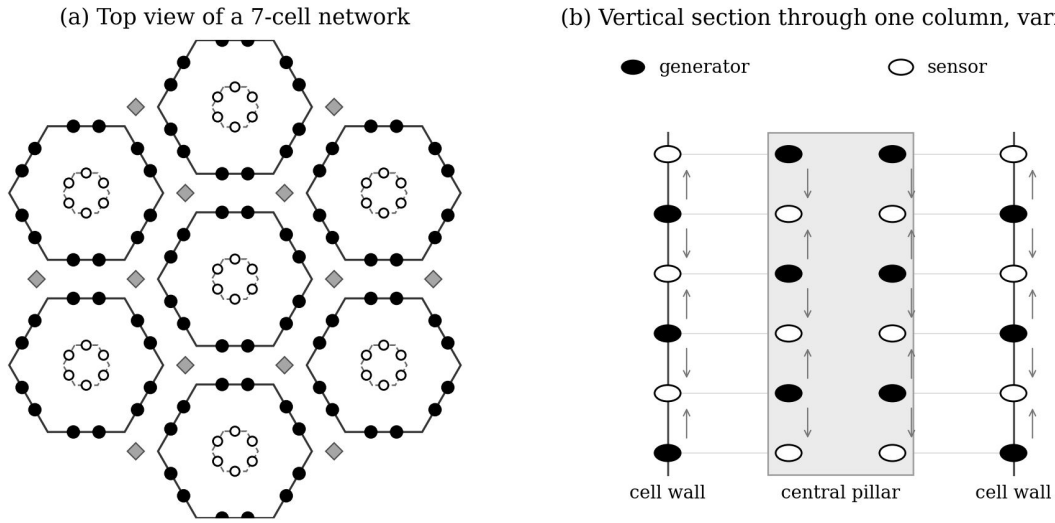


Figure 1. (a) Top view of a 7-cell network on a layer where the wall positions hold generators (filled) and the pillar positions hold sensors (open). Grey squares mark the corner channels at the junctions of three cells. (b) Vertical section through the centre of one column in variant C. The section crosses two opposite cell walls and two opposite faces of the central pillar (shaded), so each layer shows 2 of its 12 wall positions and 2 of its 6 pillar positions. Generators alternate between the walls and the pillar from layer to layer. Arrows mark the strongest couplings, from the generators directly above and below each sensor; each sensor also reads, more weakly, the other generators of its cell within one layer.

We compare four placements. In the base architecture every position is a generator with its own sensor beside it, and the generator is driven by its own sensor reading. In variant A the wall positions are generators and the pillar positions are sensors; variant B is the reverse. In variant C the roles alternate by layer: on even layers the walls are generators and the pillar is sensing, on odd layers the reverse (Fig. 1b). In A, B and C each generator receives a weighted sum of all sensor channels of its own cell within one layer above and below, which gives 13 to 25 inputs per generator, against one reading in the base architecture.

Three optional mechanisms extend the inputs. Multi-axis sensing measures three field components at two points per sensor. Frequency division assigns each generator to one of several bands and lets each sensor channel respond to one band, which multiplies the channels per sensor. Addressed long-range links give each generator inputs from the delayed states of a fixed random set of generators in other cells, routed through the corner channels.

4. Model and methods

4.1 Dynamics

Time advances in discrete steps. The effective moment of generator j is its state weakened by temperature, delayed by D steps:

$$A_j(t) = s_j(t - D) \cdot w(T), \quad w(T) = \max(0.05, 1 - c \cdot \min(1.5, T / T_{ref})) \quad (1)$$

A sensor i reads the superposed field of all generators within a cutoff radius and compresses it:

$$S_i(t) = \tanh(g \cdot \sum_j J_{ij} A_j(t) / B_s) \quad (2)$$

where J_{ij} is the field of a unit generator j at sensor i , g is a global generator strength and B_s the sensor saturation scale. In the point-dipole model J_{ij} is the axial component of the dipole field in units where the cell circumradius is 1 and $B_s = 30$. In the calibrated model J_{ij} is the field in millitesla interpolated from the MuMax3 maps of Section 4.4, with 1 model unit equal to 150 nm and $B_s = 2$ mT unless stated. Each generator relaxes towards a target drive:

$$y_i(t) = \tanh(G \cdot \sum_q W_{iq} x_q(t) - \alpha a_i(t) + L_i(t) + u_i(t) + \eta \xi_i(t)) \quad (3)$$

$$s_i(t+1) = s_i(t) + \rho [y_i(t) - s_i(t)], \quad a_i(t+1) = a_i(t) + 0.04 [s_i(t+1) - a_i(t)] \quad (4)$$

Here x_q are the sensor channels and long-range inputs of generator i with weights W_{iq} (in the base architecture the sum reduces to its own reading S_i), G is the loop gain, a_i a slow adaptation variable with strength α , L_i the input from corner logic, u_i an external input and ξ_i unit Gaussian noise of amplitude η . In reservoir experiments W is drawn once at random and kept fixed; in memory experiments it is programmed (Section 4.3).

Corner logic. Each corner channel collects a signal $v_m = 6[s_m(t - d) - s_m(t - d - 1)]$ from the generators attached to it, that is, the recent change of their state after a delay d . The output sent to a generator is $o = \tanh(\kappa \cdot \bar{v})$, where $\bar{v}$ is the mean signal from generators of the other cells at that channel and κ is the logic sensitivity; L_i is the sum of these outputs multiplied by a link strength λ .

Heat. Each generator deposits heat $q_i = h [0.15 s_i^2 g^2 + 6 |\Delta s_i| g]$ per step into a voxel grid, and each active corner channel deposits heat in proportion to its switching. Heat diffuses explicitly with reduced conductance across the gaps between cells and is removed at the corner channels, at the pillars and at the outer edge of the network. Temperature feeds back through $w(T)$ in Eq. (1). The heat model is phenomenological and serves to compare designs, not to predict absolute temperatures.

4.2 Reservoir protocol and baselines

A scalar input $u(t)$ is injected into the generators of the two lowest layers of the central cell through fixed random input weights. After a washout of 150 steps, 1400 steps are split into 800 for training, 300 for validation and 300 for testing. The features are the states of all generators and the readings of all sensors (a random subset of 500 when the network is larger) plus a constant. A linear readout is fitted by ridge regression with the regularisation chosen on the validation segment. Tasks: memory capacity $MC = \sum_k r^2(k)$ for delays $k = 1 \dots 30$ with uniform input [2]; NARMA10 [20]; Mackey–Glass prediction with delay 17 at a horizon of 5 samples, scored by normalised root-mean-square error (NRMSE) [21]; temporal XOR of the current and previous bit; parity of the last three bits; and recall of the bit two steps back. Bits are held for 4 steps and scored at the end of each bit.

Baselines are a linear readout of the raw input and echo state networks (ESN) [1, 3, 23] with 756 tanh nodes, 10% connectivity, a random bias in $[-0.5, 0.5]$ and noise of 0.01. The ESN spectral radius (0.6 or 0.95), leak rate (0.3 or 1.0) and input scale (0.5 or 1.5) were selected on seeds 1 and 2, giving 0.95, 1.0 and 1.5. We report two ESN variants: input into the same 24 nodes as the magnetic model, and input into all nodes.

Statistics. Configurations of the magnetic model were selected in a search of about 70 settings on seeds 1 and 2. Final numbers are reported on seeds 3–12 (3–8 for the calibrated model), which were not used for any selection, as mean $\pm$ 95% confidence interval from the t-distribution.

4.3 Associative memory protocol

Random patterns $\xi^\mu \in \{-1, +1\}$ over the generators are stored by programming W . For each pattern the sensor channels x^μ produced by that pattern are computed, and each generator is trained with a margin perceptron rule: whenever $\xi_i^\mu (\sum_q W_{iq} x_q^\mu + b_i) < 1$, the update $W_{iq} \leftarrow W_{iq} + 0.1 \xi_i^\mu x_q^\mu$ and $b_i \leftarrow b_i + 0.03 \xi_i^\mu$ is applied, for up to 40 passes or until all generators are fixed. Recall starts from a stored pattern with 20% or 30% of the generators flipped and runs 150 steps of Eqs. (1)–(4) with drive $\beta (\sum_q W_{iq} x_q^\mu + b_i)$ in place of the reservoir drive. The score is the overlap $m = (1/N_g) \sum_i \xi_i^\mu \text{sign}(s_i)$, where 1 is perfect recall and 0 is chance.

4.4 Micromagnetic simulations

Generators were modelled in MuMax3 [22] as perpendicular cylinders 3 nm thick with diameters of 21, 30 and 39 nm, saturation magnetisation 1.0×10^6 A/m, exchange stiffness 15 pJ/m, uniaxial anisotropy 1.0×10^6 J/m³ along z and damping 0.5, on 3 nm cells. A direct simulation of one cell of variant C over three layers (radius 150 nm, pillar radius 45 nm, layer spacing 21 nm, 30 generators) flipped each generator in turn, relaxed the system and saved the stray field. For larger networks we computed the field map of a single generator in a 405 nm cube and built every configuration by superposition, which is valid as long as the generators stay saturated. The superposition was validated against the direct 30-generator run. A sweep of 144 configurations covered 7 and 19 cells, 6 and 10 layers, layer spacings of 30, 45 and 60 nm, gaps of 45 and 75 nm, the base and C architectures and the three generator sizes. For each configuration we report the effective number of generators a sensor hears (the participation ratio of the coupling magnitudes), the share of the coupling from other layers and other cells, the median field change at a sensor between random generator patterns, the worst-case field that all other generators can apply against a generator, and the thermal stability factor $\Delta = K_{\text{eff}} V / k_B T$ at 300 K.

5. Results

5.1 Micromagnetic validation of the coupling

In the direct three-layer simulation every flipped generator remained at $m_z \approx -1$ after relaxation, so neighbouring fields did not switch any generator back. Rebuilding the same cell from single-generator maps reproduced the direct coupling matrix with a median relative error of 3.1%, 90% of the 180 significant pairs within 7.7% and a correlation of 0.99996; the strongest coupling was 25.0 mT by superposition against 24.7 mT directly. The effective dipole moment fitted to the far field agreed with the geometric moment within 1.5%, and the coupling decayed with an exponent of -2.8 , close to the dipolar -3 .

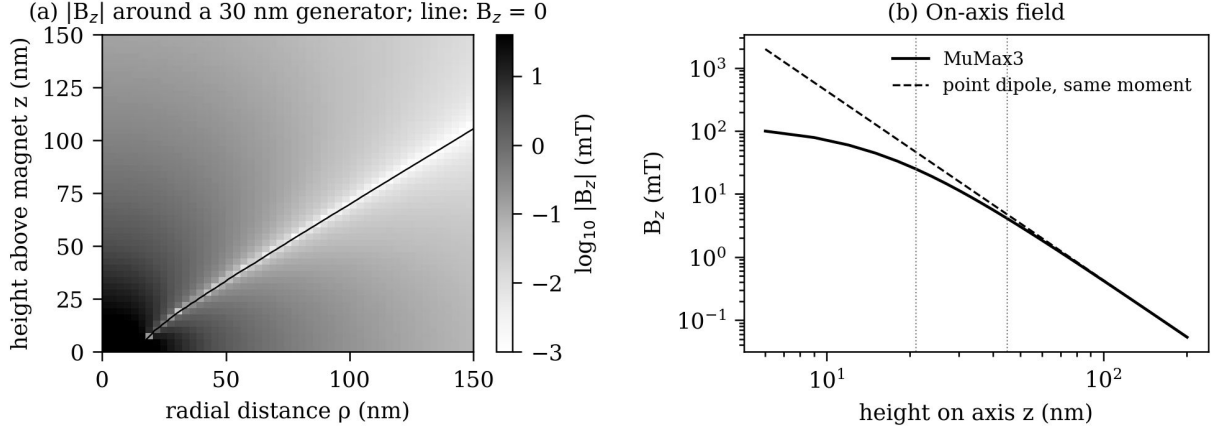


Figure 2. Stray field of one 30 nm perpendicular generator from MuMax3. (a) Axially averaged $|B_z|$ on a logarithmic grey scale; the line marks $B_z = 0$, beyond which the field reverses sign. (b) Field on the axis compared with a point dipole of the same moment. Dotted lines mark 21 nm and 45 nm.

The point-dipole approximation fails at short range (Fig. 2b): 21 nm above a 30 nm generator the field is 25 mT, against 46 mT for a point dipole of the same moment, because the moment is spread over a disk wider than the distance. The two agree beyond about 60 nm. Because the strongest couplings in a column are vertical neighbours at a few tens of nanometres, this correction changes the relative weight of near and far sources and was the main reason for calibrating the model.

Coupling in the column is local. In the 21 nm three-layer cell a sensor effectively hears 1.8 generators, the nearest one contributing 66% of the coupling and the three nearest 92%; the median coupling (0.065 mT) is about 400 times weaker than the strongest. Layer spacing controls this trade-off (Fig. 3, Table 1): in variant C with 19 cells and 10 layers, a sensor hears 3.6 generators at 30 nm and 10.8 at 60 nm, while the typical field change at the sensor falls from 21.6 mT to 2.9 mT. Between 84% and 95% of the coupling in variant C comes from other layers, which confirms that the vertical zig-zag between wall and pillar generators is the main coupling channel. At equal spacing variant C delivers three to five times stronger sensor signals than the base architecture, where each sensor hears 21 to 33 generators but each of them weakly.

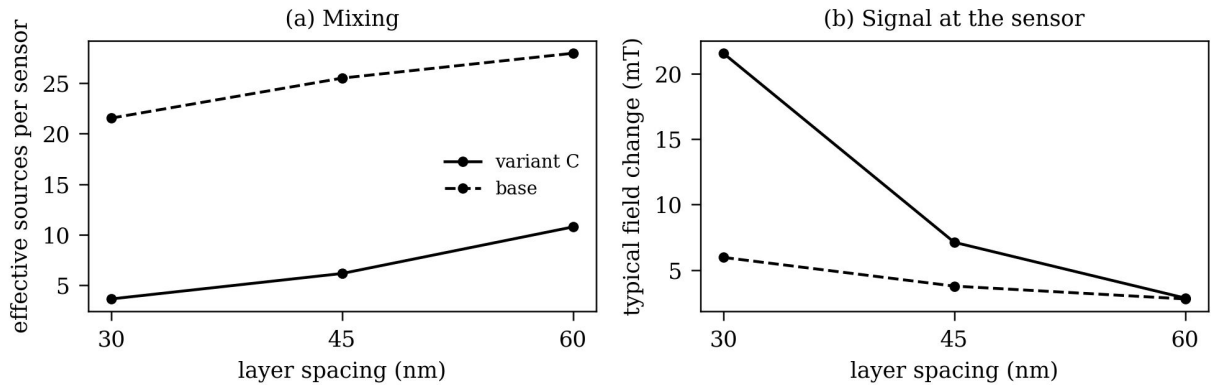


Figure 3. Effect of layer spacing in a 19-cell, 10-layer network with 30 nm generators and a 75 nm gap, from the superposition sweep. (a) Effective number of generators per sensor. (b) Median field change at a sensor between random generator patterns.

Table 1. Micromagnetic sweep for variant C, 19 cells, 10 layers, 30 nm generators. Max. neighbour = strongest single coupling from a generator of a neighbouring cell to a sensor of the central cell. Margin = effective anisotropy field (about 1000 mT) divided by the worst-case field; this is an upper bound because real nanomagnets switch below the anisotropy field.

Spacing (nm)	Gap (nm)	Sources per sensor	Nearest share	Other layers	Other cells	Max. neighbour (mT)	Field change (mT)	Worst case (mT)	Margin
30	45	3.9	0.40	0.95	0.14	0.30	21.5	27.3	36
30	75	3.6	0.41	0.95	0.11	0.24	21.6	26.1	38
45	45	7.1	0.29	0.90	0.23	0.48	6.9	20.9	48
45	75	6.2	0.31	0.90	0.17	0.08	7.1	20.0	50
60	45	13.1	0.20	0.84	0.30	0.49	2.8	17.7	56
60	75	10.8	0.23	0.85	0.22	0.05	2.9	16.9	59

Cells barely talk through the field. Other cells contribute 11–30% of the summed coupling magnitude, but this is the sum of many weak terms; the strongest single coupling from a neighbouring cell into the central cell is 0.05–0.49 mT against field changes of 3–22 mT inside the cell, and widening the gap from 45 to 75 nm reduces it by a factor of three to ten. The worst-case opposing field on a generator is 17–27 mT in variant C and up to 65 mT in the base architecture. The effective anisotropy field of the modelled material is about 1000 mT, and even if real switching fields are several times lower, the margin remains large. The thermal stability factor at 300 K is 125, 255 and 430 for the 21, 30 and 39 nm generators. Larger generators give stronger signals (10.7 mT field change at 45 nm spacing for 39 nm magnets) at the cost of a smaller margin, and 21 nm generators the reverse (3.6 mT).

5.2 Reservoir computing in the point-dipole model

Table 2 summarises ablations of the base architecture in a 7-cell, 6-layer network. Removing the field coupling, the corner logic or both coupling paths each degrades the memory capacity, and shuffling the generator positions while keeping all other parameters destroys it (MC 0.5 against 3.5) together with Mackey–Glass prediction (NRMSE 1.20 against 0.18). The geometry of the field coupling, not only the presence of many nonlinear elements, carries the computation. Heating lowered the memory capacity from 3.5 to 2.6 and pushed temporal XOR to chance.

Table 2. Ablations of the base architecture in the point-dipole model, 7 cells, 6 layers ($n = 2$ –4 seeds per row; exploratory, before the held-out protocol). The ESN row is an untuned reference without bias; the tuned baselines are in Table 3.

Configuration	MC	NARMA10 NRMSE	MG $h=5$ NRMSE	Temporal XOR	Recall $k=2$
Full model	3.5	0.68	0.18	0.63	0.81
No field coupling ($g = 0$)	2.4	0.75	0.60	0.48	0.82
No corner logic	1.5	0.77	0.31	0.60	0.70
No coupling at all	0.7	0.79	0.90	0.57	0.46
Shuffled generator positions	0.5	0.96	1.20	0.53	0.72
Untuned ESN, 756 nodes	4.0	0.65	0.20	0.53	0.80
Linear readout of input	–	0.89	0.66	0.48	–

The memory function of the full model is not monotonic (Fig. 4): it peaks at delays of 4 and 8 steps, the round-trip times of the loops formed by the corner logic delay and the generator relaxation. The network remembers in echoes rather than with a smoothly fading trace, which an ESN does not show.

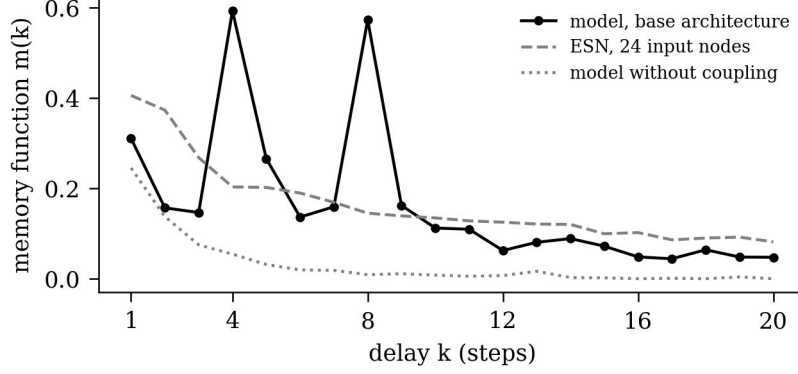


Figure 4. Memory function $m(k)$ (squared correlation between the delayed input and its reconstruction) for the base architecture, an ESN with input into 24 nodes, and the model without coupling. The peaks at $k = 4$ and 8 are loop echoes.

5.3 Placement of generators and sensors

Figure 5a compares the associative memory of the four placements with 20% damage. The base architecture, with four programmable parameters per generator (three field components and a bias), stores one pattern reliably and falls to an overlap of 0.41 at four patterns. Variants A and B hold two to four patterns before collapsing. Variant C degrades gracefully, with overlaps of 0.86, 0.83 and 0.51 at 4, 6 and 12 patterns. In reservoir tasks variant C gave the best results among the placements at low loop gain, and it uses 378 generators in a 7-cell, 6-layer network against 756 in the base architecture.

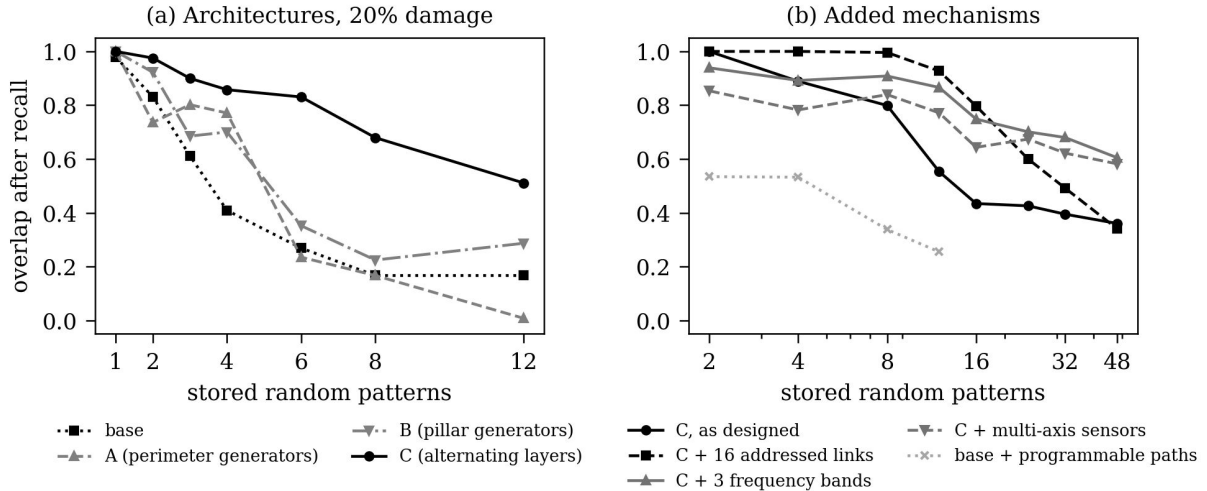


Figure 5. Associative memory after recall from 20% damage. (a) The four placements in the point-dipole model, mean of two seeds. (b) Variant C with added mechanisms, mean of two seeds; the horizontal axis is logarithmic.

5.4 Mechanisms that raise capacity

Adding inputs to each generator raises capacity in proportion to how independent those inputs are (Fig. 5b). Sixteen addressed long-range links (34 inputs per generator) recall up to 8 random patterns without error and 12 with an overlap of 0.89–0.96. Three frequency bands (54 channels) give the smoothest degradation, with an overlap of 0.57–0.64 at 48 patterns. Multi-axis sensors (108 channels) help at large pattern counts but reduce recall of few patterns, consistent with a higher sensitivity to noise. Programmable coupling paths in the base architecture, which re-route but cannot amplify the field, did not help: the field of distant generators is two to three orders of magnitude weaker than that of near ones, so the

extra paths add almost no independent information.

Increasing the number of addressed links further reaches 24 patterns with an overlap of 0.97 at 64 links and 36 patterns with 0.92 at 96 links (20% damage, one seed). Without addressed links, the best field-only option in our search was three frequency bands, with an overlap of 0.71 ± 0.03 at 24 patterns on held-out seeds, against 0.46 ± 0.03 for variant C as designed. Extending the sensor inputs to two or three layers above and below did not improve the field-only capacity.

5.5 Calibrated model against tuned baselines

Table 3 compares the selected configuration, variant C with 64 addressed links and a 75 nm gap in a 7-cell, 6-layer network, under the point-dipole model and under the MuMax3-calibrated coupling, against tuned ESNs. With the calibrated coupling the loop gain was re-selected (0.7 instead of 0.6) and a sensor scale of 2 mT was best for computing; a scale of 4 mT gave nearly perfect memory (0.99 and 0.95 at 20% and 30% damage on the selection seed) but degraded the reservoir tasks, so the sensor scale trades memory against computation.

Table 3. Held-out comparison, mean $\pm$ 95% confidence interval. MC: memory capacity ($k \leq 30$). MG: Mackey–Glass NRMSE at $h = 5$ (lower is better). Memory: overlap after recall of 24 random patterns from 20% / 30% damage.

System	Seeds	MC	MG $h=5$	Parity-3	Temporal XOR	Memory, 24 patterns
C, as designed (dipole)	10	8.6 ± 0.4	0.29 ± 0.03	0.90 ± 0.05	0.52 ± 0.04	$0.46 / 0.31$
C + 3 frequency bands (dipole)	10	8.3 ± 0.4	0.36 ± 0.01	0.90 ± 0.04	0.51 ± 0.06	$0.71 / 0.50$
C + 64 links (dipole)	10	9.4 ± 0.4	0.10 ± 0.01	0.84 ± 0.07	0.75 ± 0.07	$0.97 / 0.95$
C + 64 links (MuMax3-calibrated)	6	9.6 ± 0.6	0.091 ± 0.008	0.82 ± 0.04	0.73 ± 0.08	$0.95 / 0.89$
ESN, input to 24 nodes	10	17.9 ± 0.8	0.16 ± 0.01	0.86 ± 0.05	1.00 ± 0.00	–
ESN, input to all nodes	10	9.7 ± 0.2	0.076 ± 0.008	1.00 ± 0.00	1.00 ± 0.00	–

The calibrated model predicts the chaotic series better than the ESN that receives the same 24-node input (NRMSE 0.091 ± 0.008 against 0.16 ± 0.01), matches it on three-bit parity, and is close to an ESN that receives input into every node (0.076 ± 0.008). It is clearly weaker on memory capacity (9.6 against 17.9 for the input-matched ESN) and on temporal XOR (0.73 against 1.00). Moving from the dipole to the calibrated coupling left the reservoir results unchanged within the confidence intervals and lowered recall from 30% damage from 0.95 to 0.89.

5.6 Signal transmission between cells

We tested whether a local impulse in one cell travels to its neighbours and fades, which would allow the network to route information through physics alone. Sweeping the corner-logic sensitivity κ with a localised impulse in the central cell, we found no such regime (Fig. 6). Up to $\kappa \approx 0.65$ the response of the first ring of cells to the impulse stayed within 7% of its spontaneous activity, so the signal did not leave the cell. Between 0.70 and 0.75 the network crossed into self-sustained activity: spontaneous activity rose by a factor of two at 0.75 and of six at 0.80, and the maximum temperature rose from 1.4 to 12.7 and 16.9 model units. The transition is sharp, as expected near the edge of stability of recurrent networks [24]; in our model it is a switch between silence and a storm, not a regime of controllable propagation.

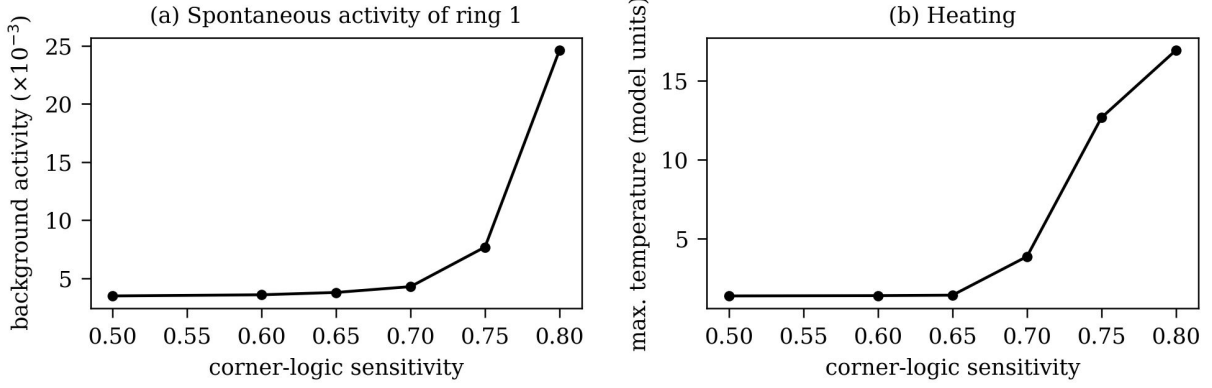


Figure 6. Inter-cell activity as a function of the corner-logic sensitivity κ in a 7-cell network. (a) Spontaneous activity of the first ring of cells. (b) Maximum temperature. Above $\kappa \approx 0.7$ activity becomes self-sustained and heating rises sharply.

6. An alternative sensor: a skyrmion film

The architecture of Sections 3–5 needs sensors that read the out-of-plane stray field and respond to it nonlinearly. The model treats them abstractly through Eq. (2), and the direct implementation is a magnetoresistive or Hall sensor at each sensor position that reads the field itself. This section examines one alternative implementation, a skyrmion in a thin film under the generators, because a skyrmion responds to a perpendicular field with a built-in nonlinearity [26–28]. It is an optional extension: the results of Sections 4 and 5 do not depend on it.

6.1 Methods

To test a physical sensor, the abstract read-out of Eq. (2) was replaced by a skyrmion in an ultrathin film placed under the generators. The film parameters follow widely used Pt/Co-like values [30]: saturation magnetisation 580 kA/m, exchange stiffness 15 pJ/m, perpendicular anisotropy 0.8 MJ/m³, damping 0.3 and interfacial DMI between 3.0 and 3.8 mJ/m². The film is 256×256 nm on 2 nm cells, with a magnetic layer 0.4, 1.2 or 2.0 nm thick treated as an effective medium for a multilayer stack [31], either continuous or patterned into a disk of 60 or 80 nm. The generator field is applied as an external field taken from the MuMax3 vector maps of Section 4.4 for 30 and 39 nm generators 21, 30 or 45 nm above the film centre. The sensor signal is the mean out-of-plane magnetisation within a disk of 20 nm radius, a proxy for the read-out of a tunnel junction above the film. Only real dynamics was used: the energy minimiser of MuMax3 failed to converge while a skyrmion was shrinking or drifting. Reference runs at 0 K used the adaptive solver. Runs at 300 K used a fixed 30 fs Heun step with the thermal field and three independent random seeds. The read accuracy is the share of read windows of 0.25 ns or 1 ns that are classified correctly by a single threshold placed midway between the pooled mean signals for generator up and generator down.

Writing was tested in two ways. A spin-polarised current through a 30 nm contact (Slonczewski torque, polarisation 0.4) was applied for 0.3 or 1 ns at 10^{11} to 3×10^{12} A/m². A gate acting through voltage-controlled magnetic anisotropy was modelled as a temporary change of the anisotropy within a 30 nm region, lowered by up to 95% for writing and raised by up to 150% for erasing, for 1 ns, with and without a generator 21 nm above that favours the target state. Gate voltage and energy were estimated for a VCMA coefficient of 100 fJ/(V·m), a 1 nm MgO barrier with relative permittivity 9.8 and the capacitance of the 30 nm gate. Each gate condition was run with three seeds and with no-pulse controls, and success was judged from the sensor signal averaged over the last 0.5 ns against reference levels measured in the same geometry for a present and an absent skyrmion. In these runs the gate region overlapped the inner part of the sensor disk, so the read-out covered the annulus between 15 and 20 nm radius, for which the reference levels were 0.60–0.72 with a skyrmion and 0.97 without.

6.2 Sensing at 0 K

At 0 K a skyrmion under the generators behaves as the sensor the architecture needs. Under three generators 45 nm above the film with four up/down patterns, the sensor signal follows the summed field at the film centre with a correlation of 0.998–0.999 for both DMI values tested, although four patterns make this a test of monotonic response rather than of linearity. The response is strongly nonlinear (Fig. 7a): for a skyrmion of 15 nm radius the signal rises by 0.28 for +20 mT, which compresses the skyrmion, and falls by 0.71 for –20 mT, which expands it, so the film provides a built-in activation function. Sensitivity depends on the size of the skyrmion relative to the read-out disk (Fig. 7b). It peaks when the skyrmion radius (18.8 nm) is close to the sensor radius (20 nm) and drops by a factor of three to four when the skyrmion grows to 28 nm and covers the sensor completely. At that optimum the contrast between generator up and down is 0.47 for a 30 nm generator 30 nm away, 0.38 for a 39 nm generator 45 nm away and 0.24 for a 30 nm generator 45 nm away. The skyrmion collapsed only at 200–480 mT and did not reappear when the field was removed, far above the 25 mT that a generator produces, so the generators cannot erase their own sensors.

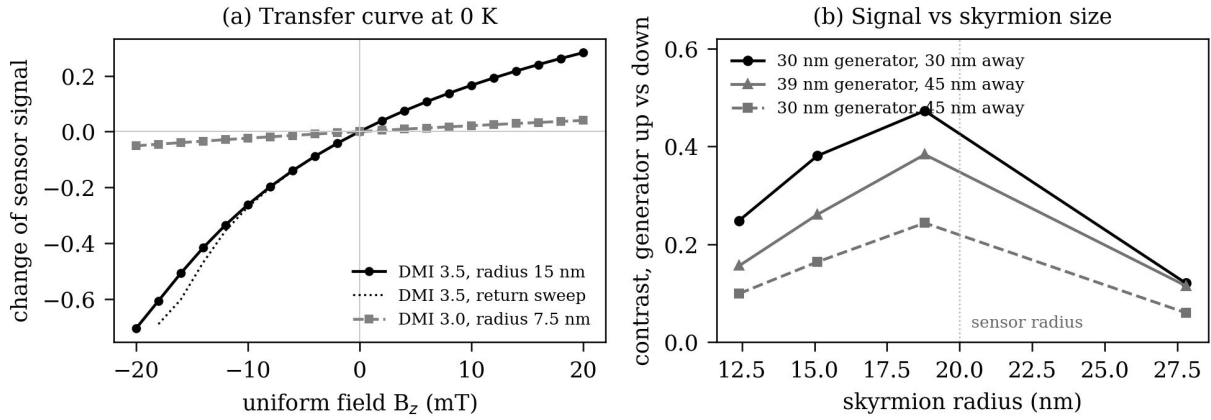


Figure 7. Skyrmion film as the sensor, 0 K. (a) Change of the sensor signal under a uniform out-of-plane field for a skyrmion of 15 nm radius (DMI 3.5 mJ/m²) and of 7.5 nm radius (DMI 3.0); the dotted line is the return branch. (b) Contrast between generator up and down as a function of skyrmion radius, set by DMI 3.4–3.7 mJ/m², for three generator configurations; the dotted line marks the read-out radius.

6.3 Sensing at 300 K

At 300 K the result depends on stability (Table 4). In an open film with a 0.4 nm magnetic layer the skyrmion wanders: the contrast varied between seeds from 0.84 to –0.12, and in one seed the sign was reversed because the skyrmion had left the sensor. Confining it in an 80 nm disk raised the read accuracy to 0.81, but with a 0.4 nm layer the skyrmion was present only 38% of the time. A thicker layer kept it alive, and with a 2.0 nm layer in an 80 nm disk and DMI 3.4 mJ/m² the skyrmion survived 100% of the time in all seeds, the contrast was 0.09–0.11 in every seed, and the read accuracy was 0.875 for 0.25 ns windows and 1.00 for 1 ns windows, that is 18 correct read-outs out of 18. Zero errors in 18 trials only bounds the error rate below about 17% at 95% confidence, so this establishes the direction but not a device error rate.

Table 4. Skyrmion sensor at 300 K, 39 nm generator 45 nm above the film, three seeds of 3 ns each. Read accuracy: share of read windows classified correctly (0.5 = guessing). Survival: worst-case share of time with the skyrmion present. A dash means the 1 ns accuracy was not computed in that run.

Geometry	Layer (nm)	DMI (mJ/m ²)	Accuracy 0.25 ns	Accuracy 1 ns	Contrast mean (min)	Survival
open film	0.4	3.6	0.69	–	0.39 (–0.12)	85%
60 nm disk	0.4	3.6	0.65	–	0.22 (0.09)	28%
80 nm disk	0.4	3.6	0.81	–	0.46 (0.15)	38%
80 nm disk	1.2	3.6	0.76	–	0.09 (0.08)	99%
80 nm disk	1.2	3.8	0.67	0.94	0.07 (0.06)	99%
80 nm disk	2.0	3.6	0.81	0.89	0.10 (0.10)	100%
80 nm disk	2.0	3.4	0.875	1.00	0.10 (0.09)	100%

6.4 Writing skyrmions

Writing skyrmions was not practical in our tests. With a current pulse, a skyrmion was written and erased only at 3×10^{12} A/m², about 2.1 mA through the 30 nm contact or 190 fJ per write at an assumed 0.3 V, one to two orders of magnitude above the current densities of commercial magnetic memory, and a generator 21 nm above did not lower this threshold on the grid tested. With a gate, nothing nucleated at 0 K at any anisotropy change, because a uniform state has nothing to break its symmetry. At 300 K in the stable 2.0 nm geometry, raising the anisotropy by 100% erased the skyrmion in 2 of 3 seeds without a generator and in 1 of 3 with one, against 0 of 3 in the controls, while lowering it by up to 95% wrote nothing. For a 2.0 nm layer these changes require about 16 V at a VCMA coefficient of 100 fJ/(V·m), so gate writing would need a material with an effect about ten times stronger. Electric-field switching of individual skyrmions has been demonstrated experimentally [32, 33], so this is a materials question rather than a physical barrier, but it is not solved here.

7. Discussion

What the results support. A column of field-coupled generator–sensor loops behaves as a recurrent computing substrate whose performance depends on its physical geometry: shuffling positions destroys it, the vertical zig-zag of variant C is the dominant coupling channel in micromagnetic simulation, and the calibrated model keeps its reservoir performance. On chaotic time-series prediction the architecture is competitive with tuned echo state networks of similar size. Single stored patterns are retained for thousands of steps in the model, binary generator states are sufficient (continuous states improved recall from 0.75 to only 0.79), and a 30% spread of element parameters is tolerable after calibration (overlap 0.82).

What they do not support. The architecture does not outperform a tuned ESN in general: its memory capacity and temporal logic are weaker. Most of the associative capacity of the best configuration comes from addressed digital links routed through the corner channels, not from the field, so that configuration is a hybrid of field-based computation inside cells and conventional routing between them. The capacity per generator follows the number of independent programmable inputs, as for a perceptron [19]: four parameters in the base architecture, 18 in variant C, 54 with frequency bands and 82 with 64 links. Closing the remaining gap to a Hopfield network of the same size, which stores about 0.138 N patterns [18], requires more independent inputs per generator, and the steep decay of the field limits how many of those inputs the field itself can provide. Field coupling between cells is too weak to carry information, and the inter-cell logic has no stable regime between isolation and runaway activity.

Open hardware questions. (1) Sensors: the calibrated configuration needs field sensors that saturate at about 2 mT and resolve fractions of a millitesla next to 30 nm magnets, and whether tunnel-junction or Hall sensors at this scale reach that resolution with acceptable noise is not addressed here. Section 6 shows that a skyrmion film confined in an 80 nm disk with a 2 nm magnetic layer is a possible alternative at 300 K, but its contrast of about 0.1 in normalised magnetisation must still be converted into an electrical signal, and a device error rate is not established. (2) Energy and speed: we did not estimate

switching energy of the generators, sensor read energy or cycle time, so no claim about efficiency relative to CMOS is made; the 1 ns read window of the skyrmion sensor corresponds to read-out at about 1 GHz. (3) The drive from sensor readings to generator switching is modelled abstractly; generators could be switched with the mechanisms of magnetic memory, but a circuit and its overhead remain to be designed. (4) The heat model is phenomenological, and the micromagnetic thermal runs use temperature-independent material parameters and small cells, which tends to overstate fluctuations. (5) Only the axial field component was calibrated with MuMax3 in the network model; lateral components still follow the dipole law scaled to the same units. (6) If skyrmion sensors are used, they do not need to be written, since the memory is held by the generators; the skyrmion writing schemes tested in Section 6 required impractical currents or voltages.

Limitations of the study. All results are from simulation. The dynamical model is phenomenological, with parameters chosen by search rather than derived from devices. The exploratory ablations (Table 2) and the mechanism comparison (Fig. 5) use two to four seeds; only Table 3 follows the held-out protocol. The associative memory benchmark uses random patterns and a perceptron programming rule; other rules and correlated patterns were not studied.

8. Conclusion

Stacked hexagonal columns of perpendicular nanomagnet generators and field sensors, coupled into closed loops through their stray fields, form a working recurrent substrate in simulation. Alternating generators between the walls and a central pillar from layer to layer gives the best memory and computing among the placements tested, micromagnetic simulation confirms that the coupling is local and dominated by the vertical neighbours, and a layer spacing near 45 nm balances mixing against signal strength. With a coupling law calibrated to MuMax3 the architecture predicts a chaotic series better than an echo state network with the same input nodes and stores 24 random patterns with an overlap of 0.95 when generators receive 64 addressed links. The same experiments show the limits of the approach: field coupling alone does not connect cells, and high associative capacity requires routed links. The next steps are a sensor noise model at the 30 nm scale, an energy and timing estimate, and micromagnetic simulation of the switching dynamics of a full column. As an optional alternative to direct field sensors, a skyrmion film can perform the field summation and the nonlinearity itself, and a skyrmion confined in an 80 nm disk with a 2 nm magnetic layer keeps reading the generator state correctly at 300 K. The field-coupled architecture does not depend on this option.

Data and code availability

The interactive simulation, the micromagnetic scripts, the MuMax3 field tables of the three generator sizes, the data of the 144-configuration sweep and the benchmark scripts that reproduce Table 3 are available at <https://github.com/veritaslinks/magnetic-columns> (interactive version: <https://veritaslinks.github.io/magnetic-columns/>). The simulation does not rely on the point-dipole approximation for the field read by the sensors: by default it interpolates the stray-field maps computed in MuMax3 for 21, 30 and 39 nm generators (Section 4.4), works in physical units (nanometres and millitesla), and a settings file loads the calibrated configuration of Table 3. The switching dynamics, the heat model and the corner logic remain phenomenological, as described in Section 4.1. The scripts and result summaries of the skyrmion sensor study of Section 6 are in the same repository.

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